

1 Möbius Inversion Formula

Theorem 1. Let f and g be two arithmetic functions satisfying for every integer $n \geq 1$:

$$g(n) = \sum_{d|n} f(d)$$

Then, for every integer $n \geq 1$:

$$f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right) = (\mu * g)(n)$$

Proof. Left as an exercise. □

2 Dirichlet Series

Definition 1. Let f be an arithmetic function and $s = \sigma + it \in \mathbb{C}$. The Dirichlet series associated to f is defined as:

$$L(f, s) = \sum_{n \geq 1} \frac{f(n)}{n^s}$$

Lemma 1. [Comparison of a Sum with an Integral] Let M, N be two real numbers $M < N$. Let $x_1, x_2, \dots, x_r$ be real numbers with $M \leq x_1 < x_2 < \dots < x_r \leq N$. Let $a(x_i)$ be a complex number for $i = 1, \dots, r$ and put:

$$A(t) = \sum_{x_i \leq t} a(x_i)$$

Let $g : [M, N] \rightarrow \mathbb{C}$ be a continuously differentiable function. Then:

$$\sum_{i=1}^r a(x_i)g(x_i) = A(N)g(N) - \int_M^N A(t)g'(t)dt$$

Proof. Put $A(x_0) = 0$ if $x_0 < M$.

$$\begin{aligned} \sum_{j=1}^r a(x_j)g(x_j) &= \sum_{j=1}^r [A(x_j) - A(x_{j-1})] g(x_j) \\ &= \sum_{j=1}^r A(x_j)g(x_j) - \sum_{j=1}^r A(x_{j-1})g(x_j) \\ &= \sum_{j=1}^r A(x_j)g(x_j) - \sum_{j=1}^{r-1} A(x_j)g(x_{j+1}) \\ &= A(x_r)g(x_r) - \sum_{j=1}^{r-1} A(x_j) [g(x_{j+1}) - g(x_j)] \end{aligned}$$

Now since $A(t) = A(x_k)$ for $x_k \leq t < x_{k+1}$, we have:

$$A(x_k) [g(x_{k+1}) - g(x_k)] = \int_{x_k}^{x_{k+1}} A(t)g'(t)dt$$

Hence:

$$\begin{aligned} \sum_{j=1}^r a(x_j)g(x_j) &= A(x_r)g(x_r) - \sum_{j=1}^{r-1} \int_{x_j}^{x_{j+1}} A(t)g'(t)dt \\ &= A(x_r)g(x_r) - \int_{x_1}^{x_r} A(t)g'(t)dt \\ &= A(N)g(x_r) - \int_M^N A(t)g'(t)dt + \int_M^{x_1} A(t)g'(t)dt + \int_{x_r}^N A(t)g'(t)dt \end{aligned}$$

- If $x_1 = M$ then $\int_M^{x_1} A(t)g'(t)dt = 0$
- If $x_1 > M$ then $A(t) = 0$, $M \leq t < x_1$ hence $\int_M^{x_1} A(t)g'(t)dt = 0$
- If $x_r = N$ we get $\int_{x_r}^N A(t)g'(t)dt = 0$ and we get the result.
- If $x_r < N$ then $A(t) = A(x_r)$ for $t \geq x_r$ hence:

$$\begin{aligned} \sum_{j=1}^r a(x_j)g(x_j) &= A(N)g(x_r) - \int_M^N A(t)g'(t)dt + A(N) \int_{x_r}^N g'(t)dt \\ &= A(N)g(N) - \int_M^N A(t)g'(t)dt \end{aligned}$$

□

Theorem 2. [Weierstrass's Convergence Theorem] Let $U \subseteq \mathbb{C}$ be a non-empty open set and $\{f_n\}$ a sequence of analytical functions $U \rightarrow \mathbb{C}$ converging pointwise to a function f on U . Assume that for every compact subset K of U , there is a constant C_K such that $|f_n(z)| < C_K$ for all $z \in K$ and $n \geq 1$. Then:

1. f is analytical on U .
2. $f_n^{(k)} \rightarrow f^{(k)}$ pointwise on U for all $k \geq 1$ and is analytical.

Theorem 3. Let f be an arithmetic function such that there exists a constant where $\sum_{n=1}^N f(n)$ is bounded for every $N \geq 1$. Then $L(s, f)$ converges for every $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 0$.

More precisely, on $\{s \in \mathbb{C} | \operatorname{Re}(s) > 0\}$, the function $L(f, s)$ is analytical and:

$$L^{(k)}(f, s) = (-1)^k \sum_{n=1}^{\infty} \frac{f(n)(\ln n)^k}{n^s}$$

Proof. On the half open plan $\operatorname{Re}(s) > 0$, we consider the partial sums:

$$\begin{aligned} L_N(s, f) &= \sum_{n=1}^N \frac{f(n)}{n^s} & N = 1, 2, \dots \\ &= \sum_{n=1}^N f(n) \exp(-s \ln n) \end{aligned}$$

Which is analytical and:

$$L_N^{(k)}(s, f) = (-1)^k \sum_{n=1}^N \frac{f(n)(\ln(n))^k}{n^s} \quad k \geq 0$$

Let $s \in \mathbb{C}$, $\operatorname{Re}(s) > 0$, using lemma 1 with $x_j = j$, $1 \leq j \leq N$ define:

$$F(t) = \sum_{1 \leq n \leq t} f(n), \quad g(t) = t^{-s}$$

Then:

$$\begin{aligned} L_N(s, f) &= F(N)N^{-s} - \int_1^N F(t)(-s)t^{-s-1}dt \\ &= F(N)N^{-s} + s \int_1^N F(t)t^{-s-1}dt \end{aligned}$$

Now taking the norm we find that the bound does not depend on N hence $L_N(s, f)$ is uniformly bounded then by theorem 2 we are done. $\square$

Corollary 1. Let f be an arithmetic function and let s_0 be a complex number such that $\sum f(n)/n^{s_0}$ is converging. Then for $s \in \mathbb{C}$ with $\operatorname{Re}(s) > \operatorname{Re}(s_0)$, the function $L(s, f)$ converges and is analytical.

Proof. Write $s = s' + s_0$ then if $\operatorname{Re}(s) > \operatorname{Re}(s_0)$ then $\operatorname{Re}(s') > 0$, we have:

$$\sum_{n \geq 1} \frac{f(n)}{n^{s'+s_0}} = \sum_{n \geq 1} \frac{f(n)}{n^{s'} n^{s_0}}$$

Then this series is bounded (why?). By applying theorem 3, we are done. $\square$

Theorem 4. There exists $\sigma(f)$ with $-\infty \leq \sigma(f) \leq \infty$ such that $L(s, f)$ converges for all $s \in \mathbb{C}$ with $\operatorname{Re}(s) > \sigma(f)$. $\sigma(f)$ is called the abscissa of convergence. Similarly, $\sigma_a(f)$ is the abscissa of absolute convergence.

Proof. Left as an exercise. □

Definition 2. The number $\sigma(|f|)$ is called the abscissa of absolute convergence of $L(s, f)$ and denoted by $\sigma_a(f)$.

Theorem 5. For any arithmetic function, we have:

$$\sigma(f) \leq \sigma_a(f) \leq \sigma(f) + 1$$

Proof. Left as an exercise. □

3 Euler Product of Dirichlet Series

Theorem 6. Let f be a multiplicative arithmetic function. If the series $L(s, f)$ converges absolutely for a complex number s , then:

$$L(f, s) = \prod_{p \text{ prime}} \left(\sum_{k=0}^{\infty} \frac{f(p^k)}{p^{ks}} \right)$$

Moreover, if f is completely multiplicative, then:

$$L(f, s) = \prod_p \left(\frac{1}{1 - f(p)p^{-s}} \right)$$

Proof. Left as an exercise. □

Example 1. For $f(n) = \mathbb{1}(n)$, $L(s, \mathbb{1}) = \zeta(s) = \sum_{n \geq 1} 1/n^s$. Since $\mathbb{1}$ is completely multiplicative:

$$\zeta(s) = \prod_p \left(\frac{1}{1 - p^{-s}} \right) \text{ for } \operatorname{Re}(s) > 1$$

4 Exercises

Exercise 1. Write the following Dirichlet series in terms of the Riemann Zeta function $\zeta(s)$:

$$1. \sum_{n \geq 1} \frac{\tau(n^2)}{n^s}$$

$$2. \sum_{n \geq 1} \frac{\omega(n)}{n^s}$$

$$3. \sum_{n \geq 1} \frac{2^{\omega(n)}}{n^s}$$

$$4. \sum_{n \geq 1} \frac{2^{\omega(n)} \lambda(n)}{n^s}$$

$$5. \sum_{n \geq 1} \frac{\kappa(n)}{n^s}$$

$$6. \sum_{n \geq 1} \frac{3^{\omega(n)} \kappa(n)}{n^s}$$

$$7. \sum_{n \geq 1} \frac{3^{\omega(n)} \kappa(n) \lambda(n)}{n^s}$$

Where:

$$\kappa(n) = \begin{cases} 1 & \text{if } n = 1 \\ a_1 \cdots a_k & \text{if } n = p_1^{a_1} \cdots p_k^{a_k} \end{cases}$$

Exercise 2. If $f(n)$ is a completely multiplicative arithmetic function prove that:

$$\frac{L'(f, s)}{L(f, s)} = - \sum_{n \geq 1} \frac{f(n) \Lambda(n)}{n^s}$$